

Exercise 1: Mocking and Stubbing

ExternalAPI.java

```
package junitdemo;

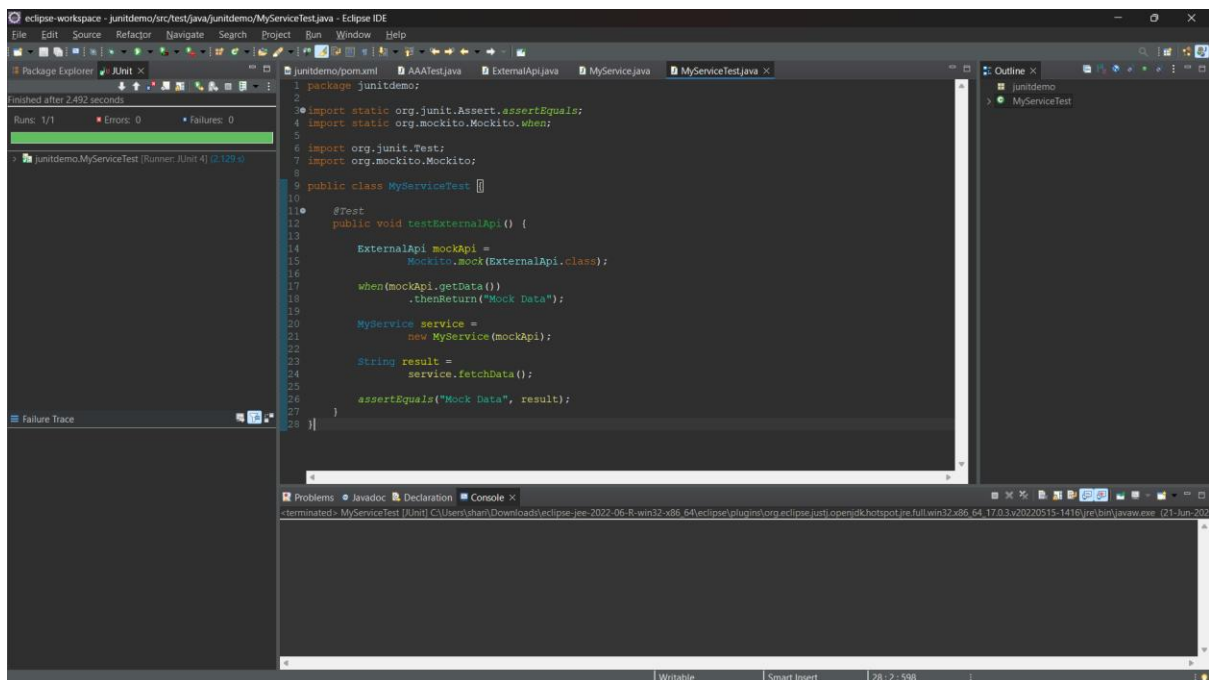
public interface ExternalApi {
    String getData();
}
```

MyService.java

```
package junitdemo;

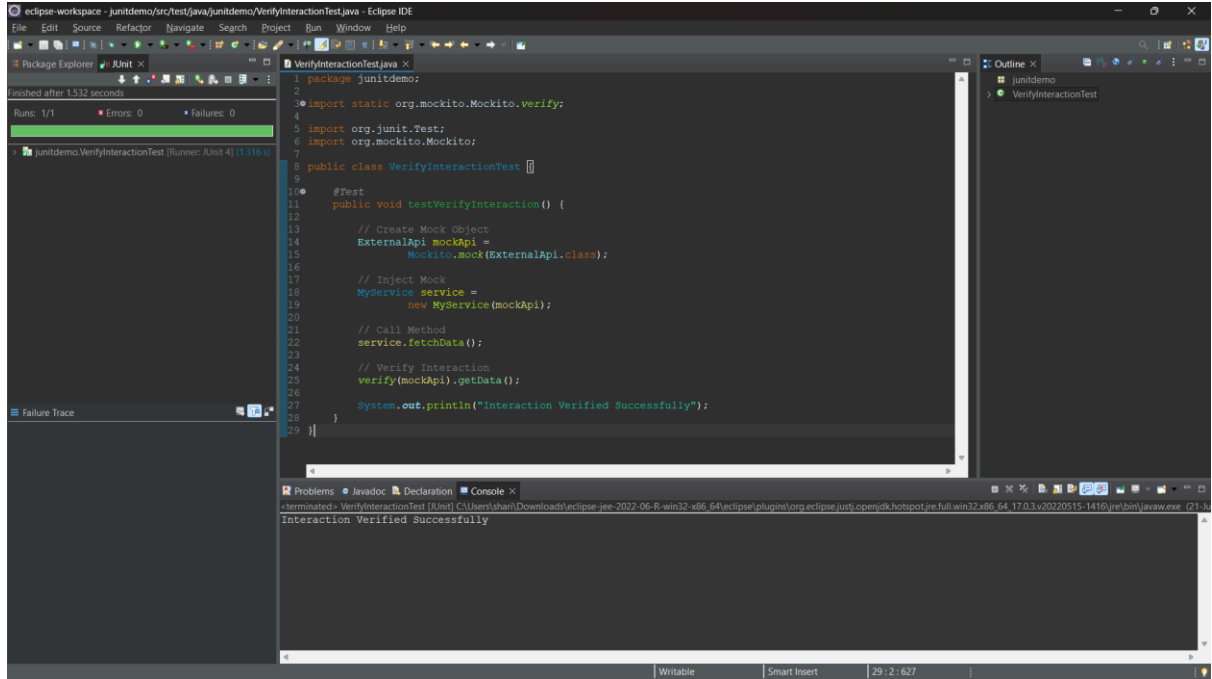
public class MyService {
    private ExternalApi api;
    public MyService(ExternalApi api) {
        this.api = api;
    }
    public String fetchData() {
        return api.getData();
    }
}
```

OUTPUT



Exercise 2: Verifying Interactions

OUTPUT



The screenshot displays the Eclipse IDE interface with the following components:

- Package Explorer:** Shows the project structure with 'JUnit' selected.
- JUnit Runner:** Indicates 'JUnit 4' is used, with a run time of 1.532 seconds, 0 errors, and 0 failures.
- Source Editor:** Contains the code for `VerifyInteractionTest.java`. The code imports `org.mockito.Mockito.verify`, `org.junit.Test`, and `org.mockito.Mockito`. It defines a `VerifyInteractionTest` class with a `@Test` method `testVerifyInteraction()`. Inside this method, it creates a mock `ExternalApi`, injects it into a `MyService` instance, calls `service.fetchData()`, and then verifies the interaction using `verify(mockApi).getData()`. A success message is printed to the console.
- Outline:** Lists the `VerifyInteractionTest` class.
- Console:** Shows the output of the test run, which is 'Interaction Verified Successfully'.

```
1 package junitdemo;
2
3 import static org.mockito.Mockito.verify;
4
5 import org.junit.Test;
6 import org.mockito.Mockito;
7
8 public class VerifyInteractionTest {
9
10     @Test
11     public void testVerifyInteraction() {
12
13         // Create Mock Object
14         ExternalApi mockApi =
15             Mockito.mock(ExternalApi.class);
16
17         // Inject Mock
18         MyService service =
19             new MyService(mockApi);
20
21         // Call Method
22         service.fetchData();
23
24         // Verify Interaction
25         verify(mockApi).getData();
26
27         System.out.println("Interaction Verified Successfully");
28     }
29 }
```

Interaction Verified Successfully